

HH is the baseline because water cancels from the ideal ratio

Acetate-buffer model

$$C_{HA} = \frac{C_0 F_H}{F_H + F_A + F_W}, \quad C_{A^-} = \frac{C_0 F_A}{F_H + F_A + F_W}$$

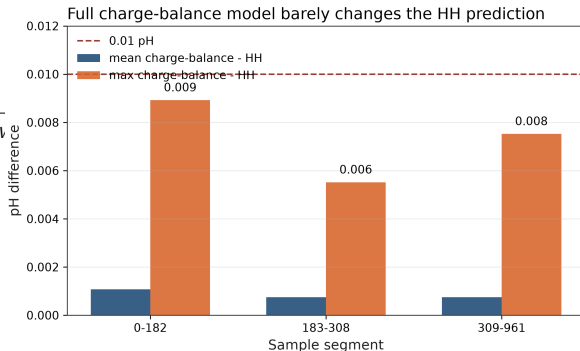
$$\text{pH} \approx \text{p}K_a + \log_{10} \frac{C_{A^-}}{C_{HA}} \approx \text{p}K_a + \log_{10} \frac{F_A}{F_H}$$

Small interpretation

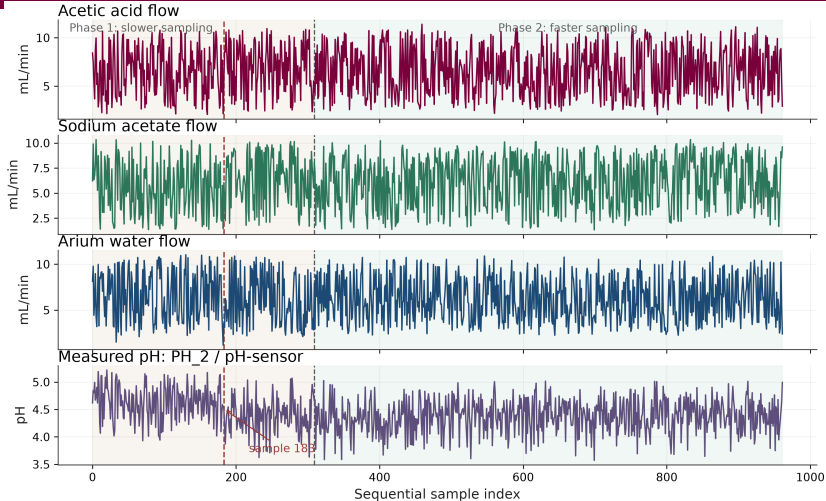
Water should not be forced to explain steady-state pH. It belongs in dilution, buffer strength, total flow, delay, and sensor-response terms.

Data check

Charge-balance equilibrium with water changes HH by less than 0.01 pH in the checked data.

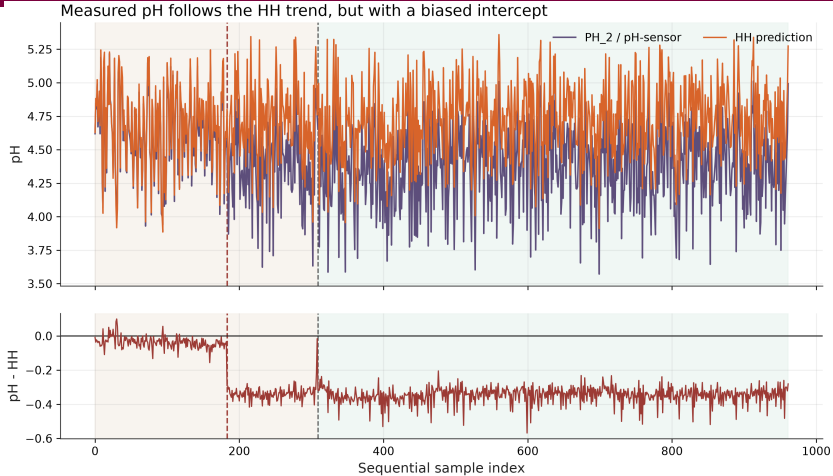


Updated dataset: sequential view of flows and PH_2



Small interpretation: x-axis is sample index. Sample 183 is the main residual-shift boundary; sample 309 is only the sampling-rate phase change.

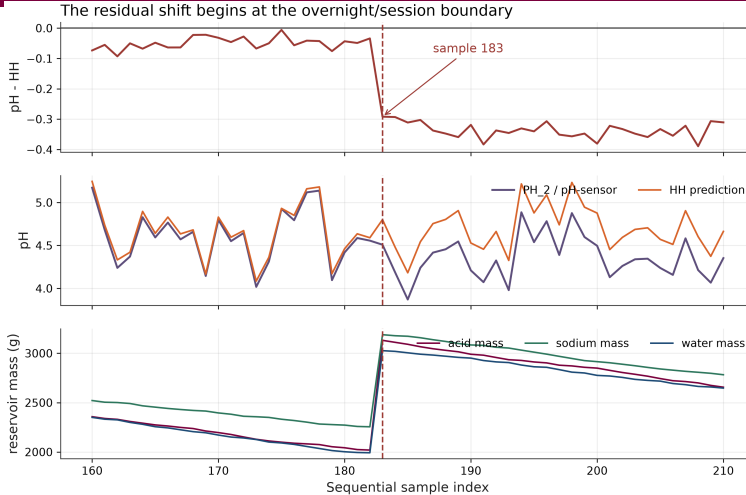
HH prediction tracks the trend, but the residual becomes biased



Overall: mean error = -0.286 , MAE = 0.287 , RMSE = 0.314 , correlation = 0.913 .

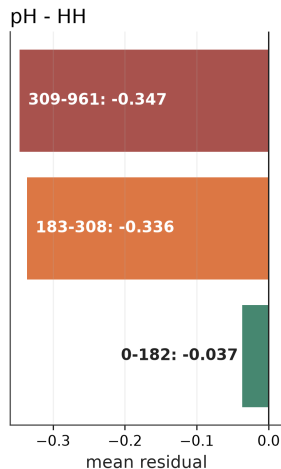
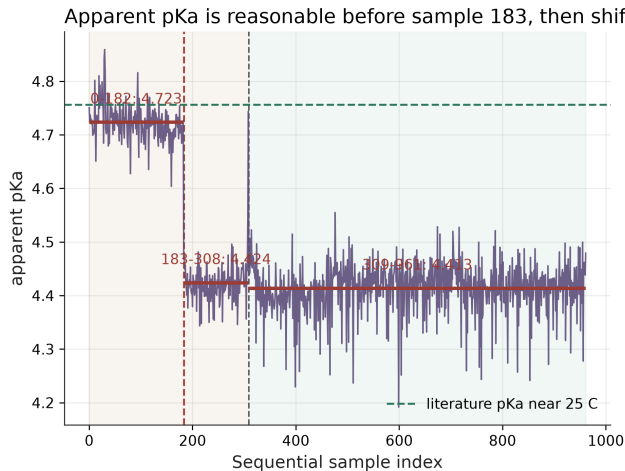
Bias is an intercept problem more than a ratio-trend problem.

The residual shift occurs at the overnight/session boundary



Small interpretation: after sample 183, PH_2 remains below the HH prediction while reservoir masses reset upward.

Apparent pKa confirms a regime-dependent intercept shift



$$pK_{a,app,k} = \text{pH}_k - \log_{10}(F_{\text{acetate},k}/F_{\text{acid},k})$$

Post-183 value is not a normal temperature pKa.

Raw three-flow control is non-unique for pH tracking

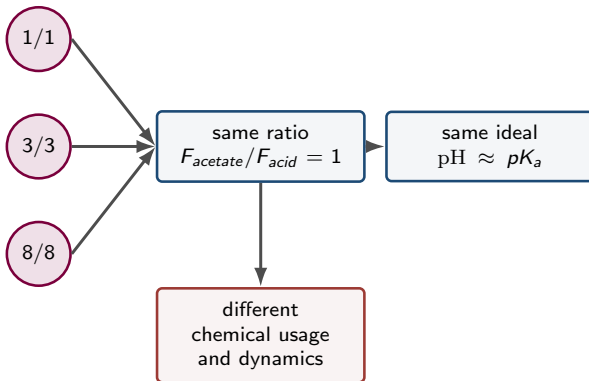
Problem with raw actions

$$u = \begin{bmatrix} F_{acid} \\ F_{acetate} \\ F_W \end{bmatrix}, \quad \text{pH} \approx \text{p}K_a + \log_{10} \frac{F_{acetate}}{F_{acid}}$$

$$(1, 1), (3, 3), (8, 8) \Rightarrow F_{acetate}/F_{acid} = 1$$

Document warning

- drifting to unnecessarily high flows,
- drifting to very low flows,
- unstable/chattering pump actions,
- meaningless use of water or total flow.



The pH reward alone cannot choose the “best” flow pair.

Controller structure 1: ratio-based controller

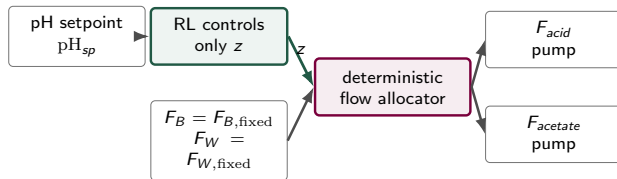
Reparameterize the action

$$r = \frac{F_{\text{acetate}}}{F_{\text{acid}}}, \quad z = \log_{10} \frac{F_{\text{acetate}}}{F_{\text{acid}}}$$

$$\text{pH} \approx \text{p}K_a + z, \quad F_B = F_{\text{acid}} + F_{\text{acetate}}$$

Deterministic flow allocator

$$r = 10^z, \quad F_{\text{acid}} = \frac{F_B}{1+r}, \quad F_{\text{acetate}} = \frac{rF_B}{1+r}$$



First RL version: RL controls only z ; F_B and F_W are fixed or chosen by a simple rule.

Controller structures 2–3: correction and usage objective

HH baseline + RL correction

$$z_{HH} = \text{pH}_{sp} - \text{p}K_a, \quad z_{RL} = z_{HH} + \Delta z_{RL}$$

$$r = 10^{z_{RL}}, \quad F_{acid} = \frac{F_B}{1+r}, \quad F_{acetate} = \frac{rF_B}{1+r}$$

Δz_{RL} corrects

- sensor bias,
- mixing delay,
- activity effects,
- model mismatch.

Chemical usage objective

$$R_k = -Q(\text{pH}_k - \text{pH}_{sp})^2 - \lambda_{\Delta u} \|\Delta u_k\|^2 \\ - \lambda_B (F_B - F_{B,ref})^2 - \lambda_{chem} (F_{acid} + F_{acetate})$$

$$C_{buffer} \geq C_{min} \quad C_{buffer} = \frac{C_0(F_{acid} + F_{acetate})}{F_{acid} + F_{acetate} + F_W}$$

Clean hierarchy from the document

pH tracking	control acid/acetate ratio
buffer strength	control total acid+acetate flow
dilution/residence time	control water or total flow